DisplayCAL-CG

Display calibration and characterization powered by ArgyllCMS

App version 3.10.0.dev82

English edition

DisplayCAL-CG User Guide

Monitor calibration and profiling, step by step — GDC edition

Document version 1.0 · based on DisplayCAL-CG 3.10.0.dev82

Contents

1. What is DisplayCAL-CG

DisplayCAL-CG is a GDC edition of the open-source DisplayCAL project (the community continuation of Florian Höch's original work), repackaged and fully translated by GDC. The application calibrates and profiles your monitor using the ArgyllCMS measurement engine — the result is a monitor with correct, consistent colors no matter which application you're working in (photo, video, graphics).

> **Calibration vs. profiling — what's the difference**
> Calibration brings the monitor to a known target state (white point, luminance, tone curve) through hardware/software adjustments (the video card's curves). Profiling then measures how the calibrated monitor responds and writes an ICC profile that the rest of the system uses for accurate color correction. They're always done one after the other, in this order.

This guide walks through each of the 5 main tabs (Display & instrument, Calibration, Profiling, 3D LUT, Verification), the advanced tools available from the menus, and how to install/update the application on Mac and Windows.

> **You need a measurement device**
> DisplayCAL-CG CANNOT calibrate a monitor without a physical colorimeter or spectrophotometer connected via USB (e.g. X-Rite i1Display Pro, ColorMunki Display, Datacolor Spyder). The app auto-detects the connected device on the "Display & instrument" tab.

2. Installation

macOS

1. Download the DisplayCAL-CG.pkg package from the download page (gordas.dev/DisplayCAL-CG).
2. Double-click the downloaded .pkg file.
3. In the installer window, click Continue on each step until you reach the License page.
4. Read the GPLv3 license summary shown, click Agree (without explicit acceptance, the install cannot proceed — this is a requirement of the native macOS installer, not an optional step).
5. Click Install — the app and all 9 satellite tools (Profile Info, Curve Viewer, etc.) are installed directly into Applications, nothing to drag manually.
6. Open DisplayCAL-CG from Launchpad/Applications.

> **The package is Apple-signed and notarized**
> macOS Gatekeeper will let the app launch directly, without the "cannot be opened because it is from an unidentified developer" warning.

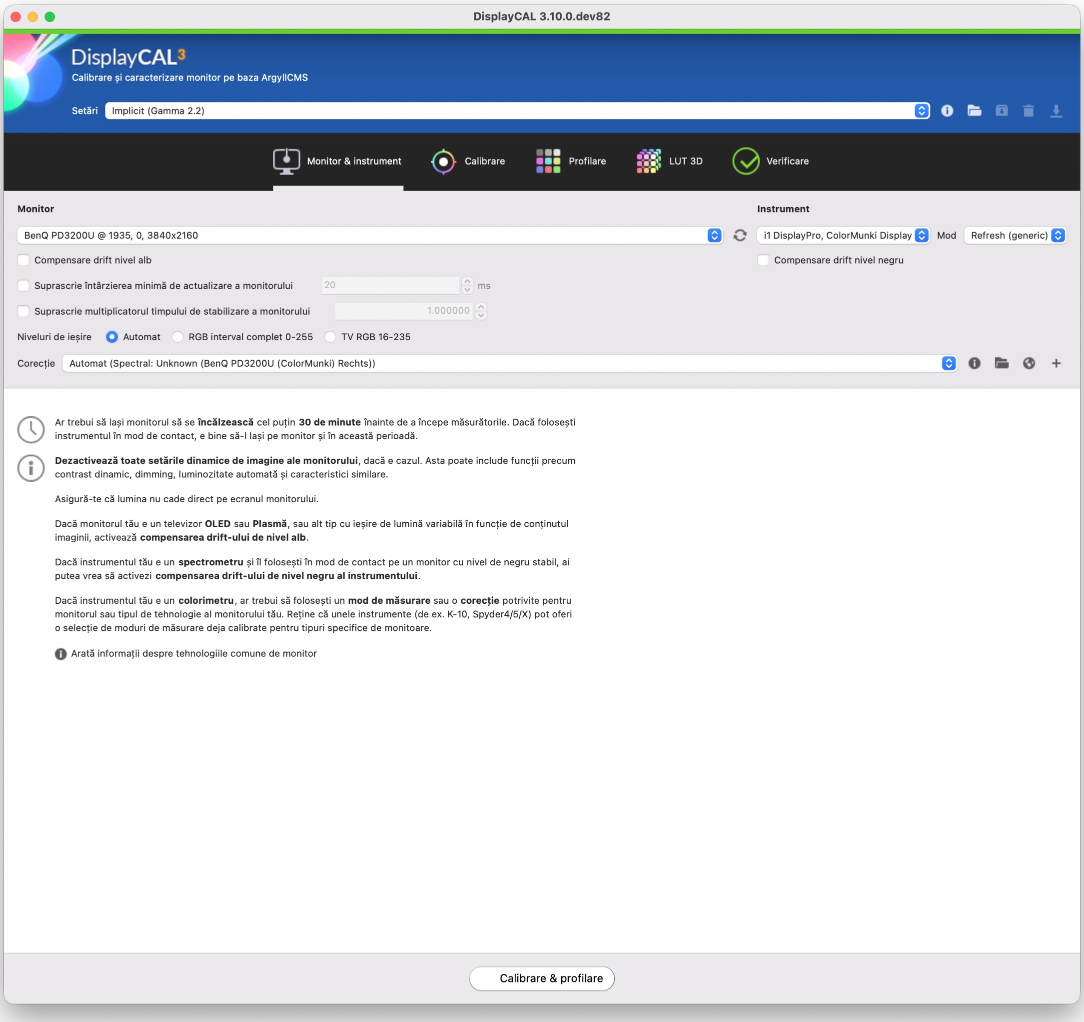
Windows

1. Download the DisplayCAL-CG-Setup.exe installer from the download page.
2. Double-click the downloaded file.
3. If the Windows SmartScreen warning appears ("Windows protected your PC"), click More info, then Run anyway.
4. On the installer's license page, select I accept the agreement — the "Next" button stays disabled until you check this.
5. Follow the installer's steps (the default location is recommended) through to Install.
6. Open DisplayCAL-CG from the Start Menu or the Desktop shortcut.

Why the SmartScreen warning appears
The Windows installer isn't (yet) signed with a paid code-signing certificate — the warning is normal for unsigned software, not a sign that the file is unsafe. Only download it from the official page above.

3. Display & instrument

The first tab — pick WHICH monitor you're calibrating and WITH WHICH instrument. The app needs to know both before you can move on to Calibration.



The Display & instrument tab, with an external monitor and an i1DisplayPro colorimeter already detected.

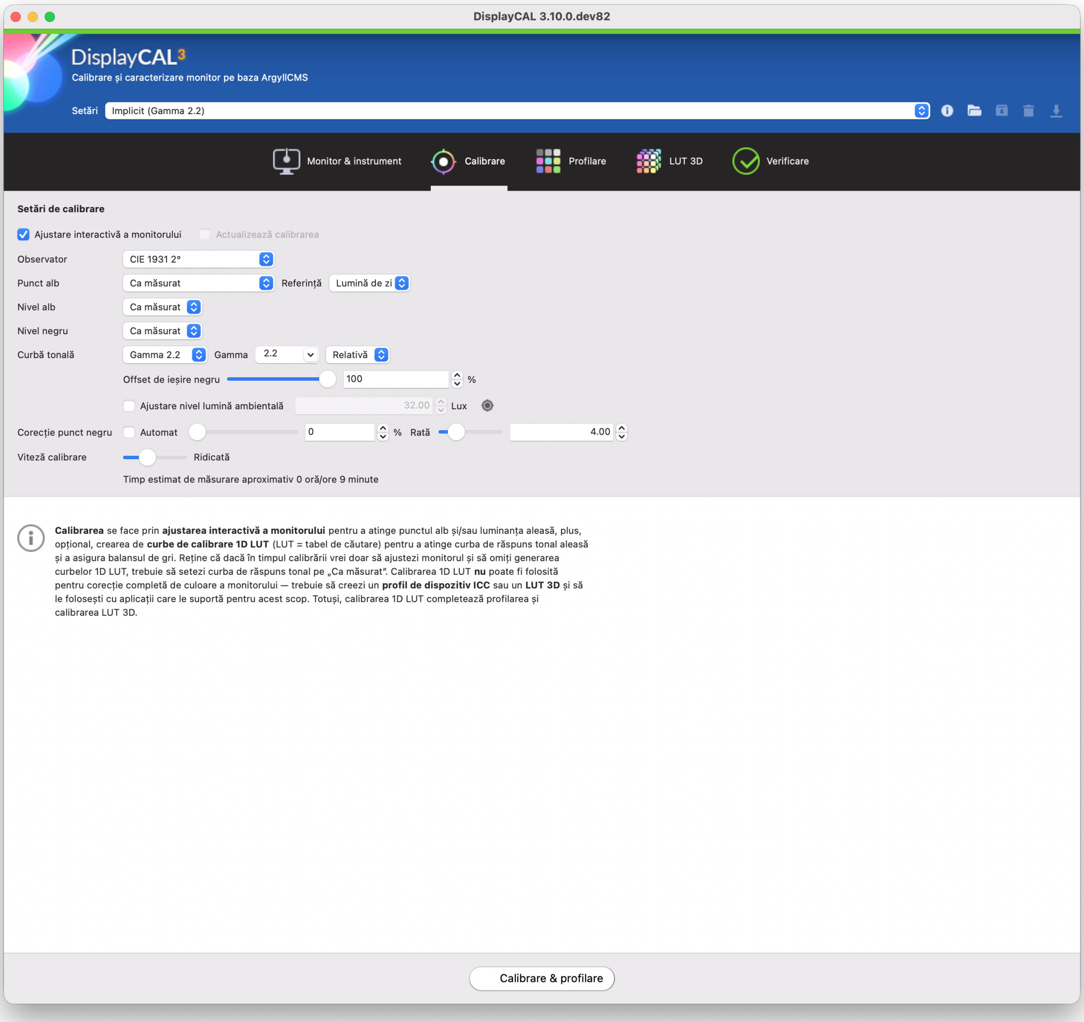
Field	What it does
Display	Pick the monitor to calibrate from the list if you have more than one connected. The round button next to it rescans connected displays.
Instrument	The colorimeter/spectrophotometer detected via USB. If it's empty, check the USB cable and that the instrument is recognized by the operating system.
Measurement mode	The instrument's measurement mode — "Refresh (generic)" suits most LCD/LED monitors. Some instruments (K-10, Spyder4/5/X) offer precalibrated modes for specific display types — pick the closest match if one exists.
White level drift compensation	Enable if the display is an OLED/Plasma TV or another type with variable light output depending on on-screen content.
Black level drift compensation	Enable if you're using a spectrometer in contact mode on a display with an unstable black level.

Field	What it does
Output levels	"Auto" is correct in nearly all cases. "TV RGB 16-235" only applies if the display/video card deliberately limits the signal range, as with a TV used as a monitor.
Correction	The instrument+display specific color correction — DisplayCAL-CG picks it automatically ("Auto (Spectral: ...)") when it can; don't change it manually unless you know exactly what you're doing.

> **Before you measure**
> Let the display warm up for at least 30 minutes before calibrating — a cold monitor's colors drift as it stabilizes thermally. Turn off any dynamic picture settings (dynamic contrast, auto brightness) and avoid light falling directly on the screen during measurement.

4. Calibration

The second tab — choose WHAT target state you want for the display: what white point, what luminance, what tone curve.



Default calibration settings (Gamma 2.2, white point and levels "as measured").

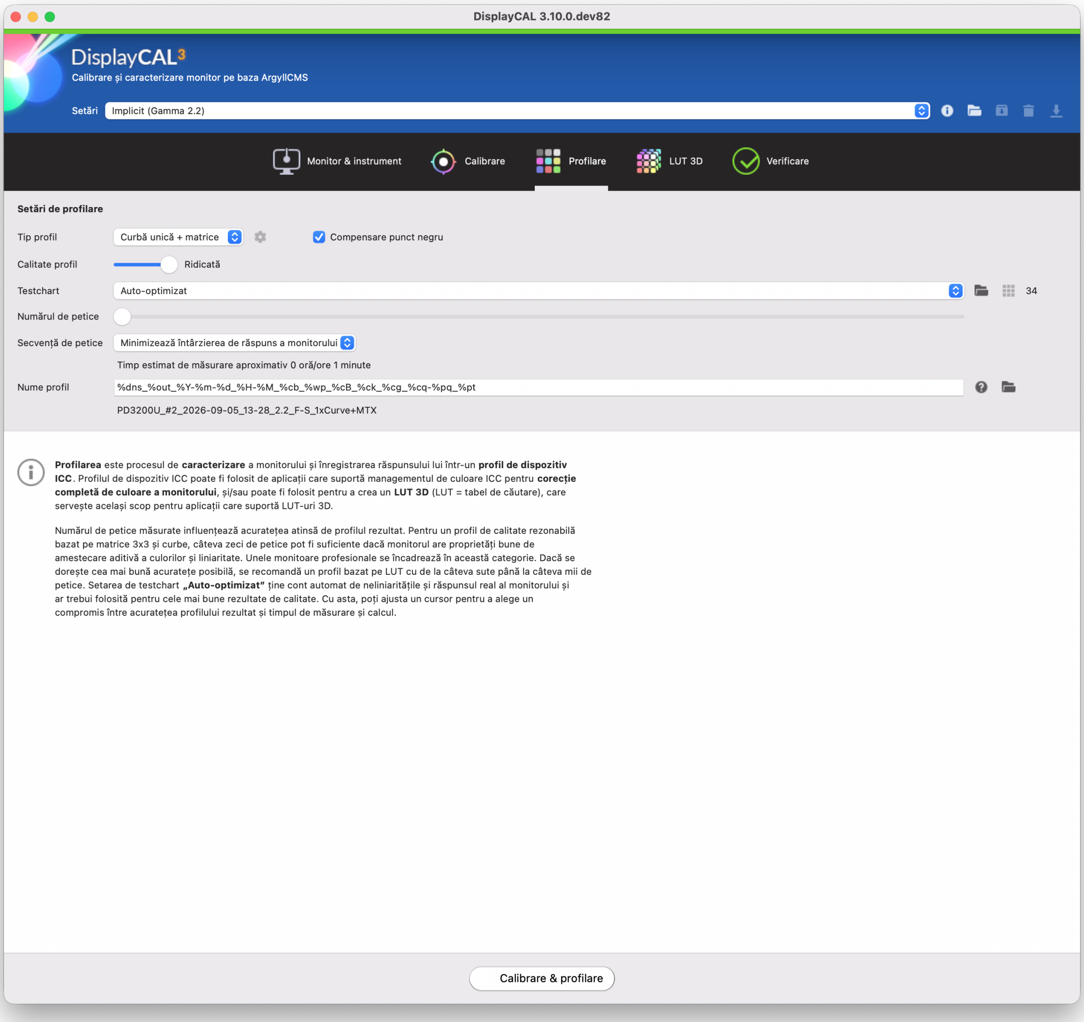
Field	What it does
Interactive display adjustment	When checked, the app guides you to manually adjust the monitor's physical controls (brightness, contrast, RGB) during calibration, to get as close as possible to the target before generating the software curves.
Observer	The CIE standard used to interpret color — "CIE 1931 2°" is the default, suitable for the vast majority of situations.
White point	"As measured" keeps the display's native white point. You can instead pick a fixed color temperature (e.g. 6500K/D65) if you need to hit an exact standard, with a reference ("Daylight"/"Black body").
White level / Black level	"As measured" keeps the display's native luminance. You can set a fixed value manually (e.g. 120 cd/m²) if you need to meet a luminance standard.
Tonal response curve	The shape the resulting tone response will have — "Gamma 2.2" is the default fit for photo/web; "Rec. 1886" or other curves matter mostly for video.

Field	What it does
Black output offset	0% = "pure" black; 100% = black follows the chosen curve exactly with no offset. In-between values compensate for displays with elevated black levels.
Black point correction	The rate/percentage used to correct near-black nonlinearities — "Auto" lets the app decide.
Calibration speed	A time/accuracy trade-off — "High" (default) is enough for most uses.

1D LUT calibration doesn't replace an ICC profile
The curves generated here only correct the display's overall tonality — for full color correction you also need a device ICC profile or a 3D LUT, created in the following tabs.

5. Profiling

The third tab — DisplayCAL-CG displays actual color patches on screen, measures them with the instrument, and builds the ICC profile that characterizes your now-calibrated display.



Profiling settings — "Single curve + matrix" profile type, auto-optimized testchart, 34 patches.

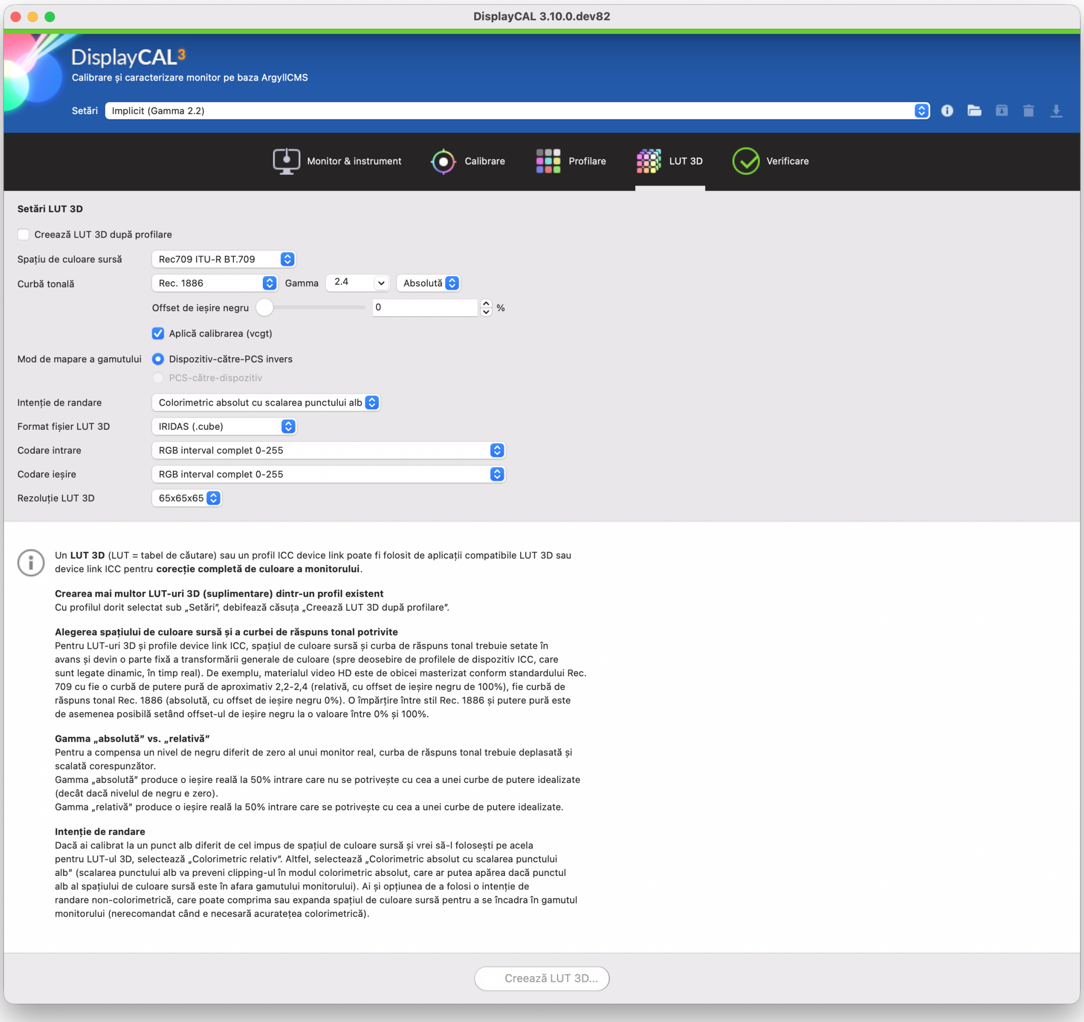
Field	What it does
Profile type	"Single curve + matrix" is fast and sufficient for many good displays; a LUT-based profile with hundreds to thousands of patches gives the best possible accuracy, but takes much longer.
Black point compensation	Recommended checked — improves accuracy in dark areas.
Profile quality	Low→High slider — affects how finely the profile is computed from the measured data, not the number of patches.
Testchart	"Auto-optimized" automatically picks patch distribution accounting for your display's actual nonlinearities — recommended for the best results.
Number of patches	The slider controls how many patches get measured — more patches = a more accurate profile, but a longer measurement.
Patch order	"Minimize display response delay" orders the patches to shorten total measurement time.

Field	What it does
Profile name	The automatic naming template — you can freely edit the field below if you want a custom profile name.

> **How long it takes**
> The app shows an estimated time under the profiling settings (e.g. "about 1 minute" for 34 patches) — a LUT-based profile with thousands of patches can take anywhere from a few minutes to over an hour.

6. 3D LUT

An optional tab — generates a 3D LUT (a three-dimensional lookup table) from the profile already created, for applications that support color correction via 3D LUT instead of an ICC profile (common in video/color-grading workflows — DaVinci Resolve, media players).



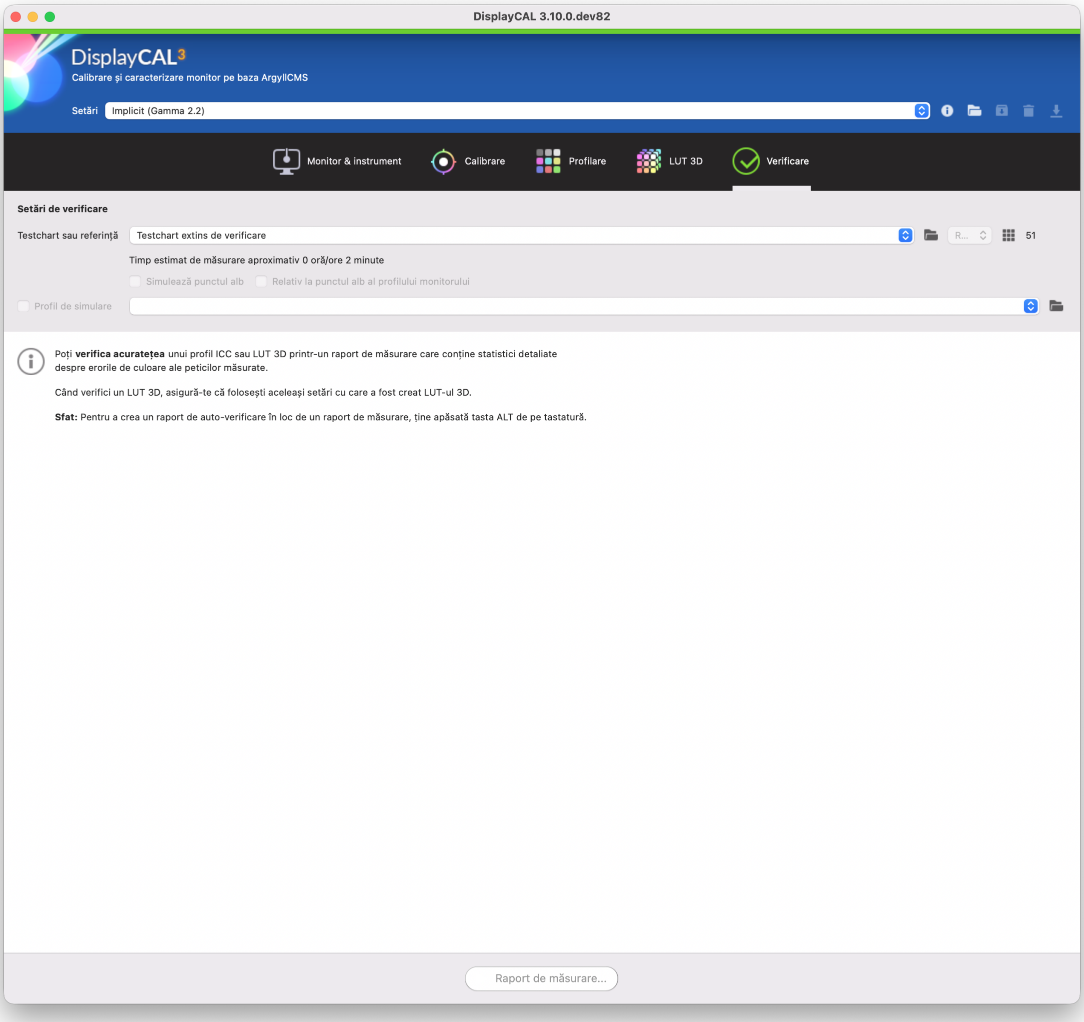
3D LUT settings — Rec709 source, Rec.1886 curve, IRIDAS .cube format, 65×65×65 resolution.

Field	What it does
Create 3D LUT after profiling	Check this if you want the LUT to be generated automatically right after profiling finishes, as one continuous step.
Source colorspace	The colorspace of the material you'll be playing back (e.g. "Rec709 ITU-R BT.709" for standard HD video).
Tone curve	Must match the source material's standard — HD video typically uses either a ~2.2-2.4 power curve or "Rec. 1886".
Gamut mapping mode	"Inverse device-to-PCS" is the standard choice for a display LUT (not a content-conversion LUT).
Rendering intent	"Absolute colorimetric with white point scaling" is recommended if you haven't explicitly calibrated to the source material's white point.
3D LUT file format	IRIDAS .cube is the most widely compatible (Resolve, most media players); other formats exist for specific software.
3D LUT resolution	65×65×65 is a good accuracy/file-size trade-off — higher resolutions increase both precision and file size.

Use the SAME settings the LUT was created with
When later verifying an already-created 3D LUT (Verification tab), make sure you use exactly the same settings (source space, curve, rendering intent) — otherwise the verification result is meaningless.

7. Verification

The fifth tab — checks how accurate an already created ICC profile or 3D LUT is, via a measurement report with statistics on the color errors measured across a set of patches.



Verification settings — extended verification testchart, 51 patches, ~2 minutes estimated.

Field	What it does
Testchart or reference	The patch set used for verification — "Extended verification testchart" is a standard set, independent of the one used for profiling (otherwise the verification would be biased).
Simulate white point	Compares the result relative to a simulated white point, instead of the display's native one.
Relative to display profile white point	Similar to the above, but relative to the white point RECORDED in the current profile.
Simulation profile	Optional — checks how the display would behave if it simulated another profile/colorspace.

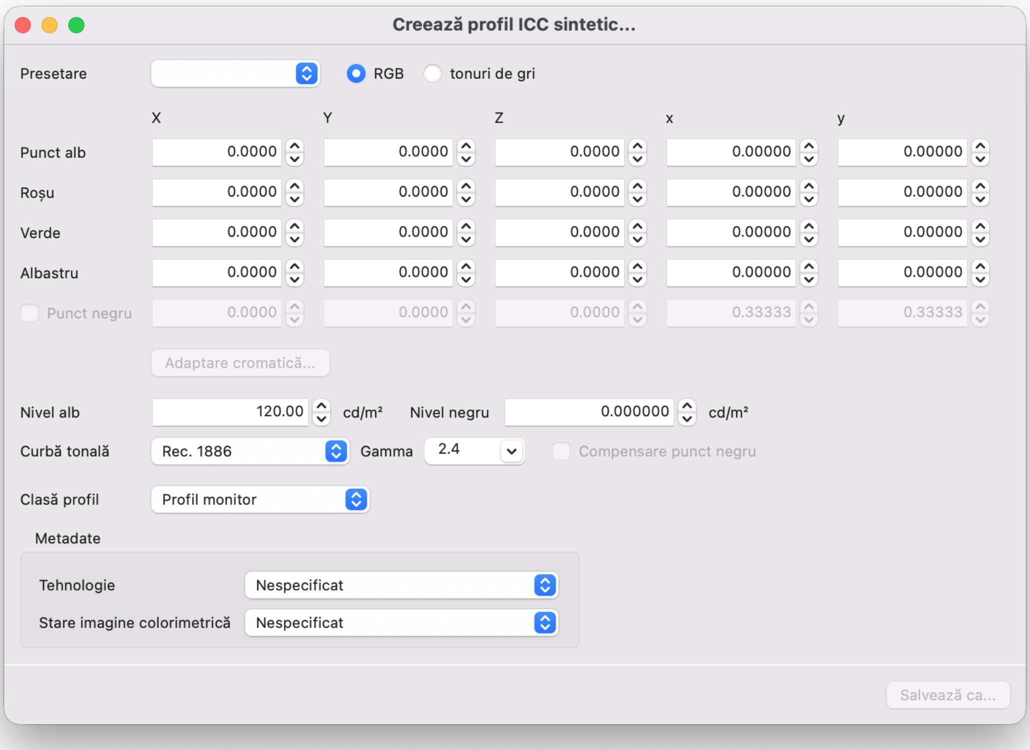
1. Pick the verification testchart (the default suits almost every case).
2. Click Measurement report... at the bottom of the window.
3. Follow the instrument on screen as the app displays and measures the test patches one by one.
4. At the end, a report opens with the average/maximum color errors measured (ΔE) — the lower the ΔE, the more accurate the profile.

> **Tip**
> Hold down the ALT key on your keyboard when clicking "Measurement report..." to create a self-check report instead of a regular measurement report.

8. Advanced tools

Besides the main workflow (the 5 tabs above), DisplayCAL-CG includes several standalone tools, useful for special cases — accessible from the main app's menu or as separate applications installed alongside it.

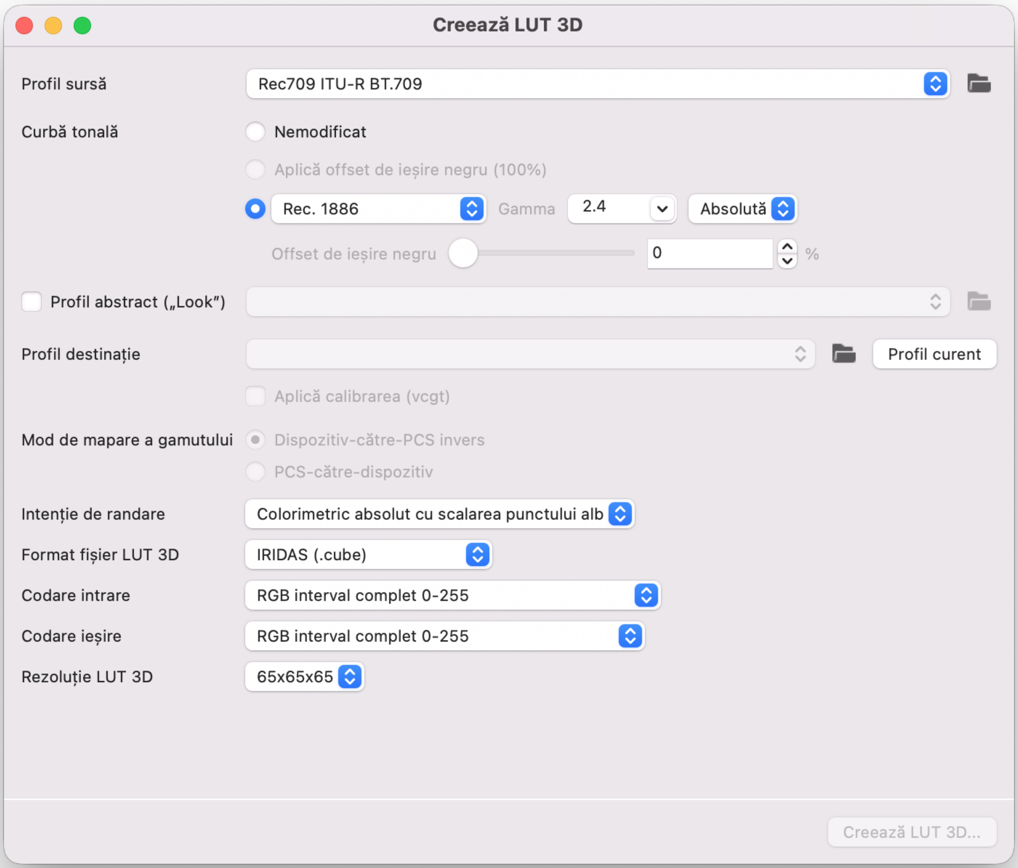
Create synthetic ICC profile



The synthetic ICC profile creation tool, built from manually described parameters (not from real measurements).

Builds an ICC profile from manually described parameters (white point, gamma, color primaries) — useful for generating a theoretical reference profile without measuring a real display (e.g. for simulation or testing).

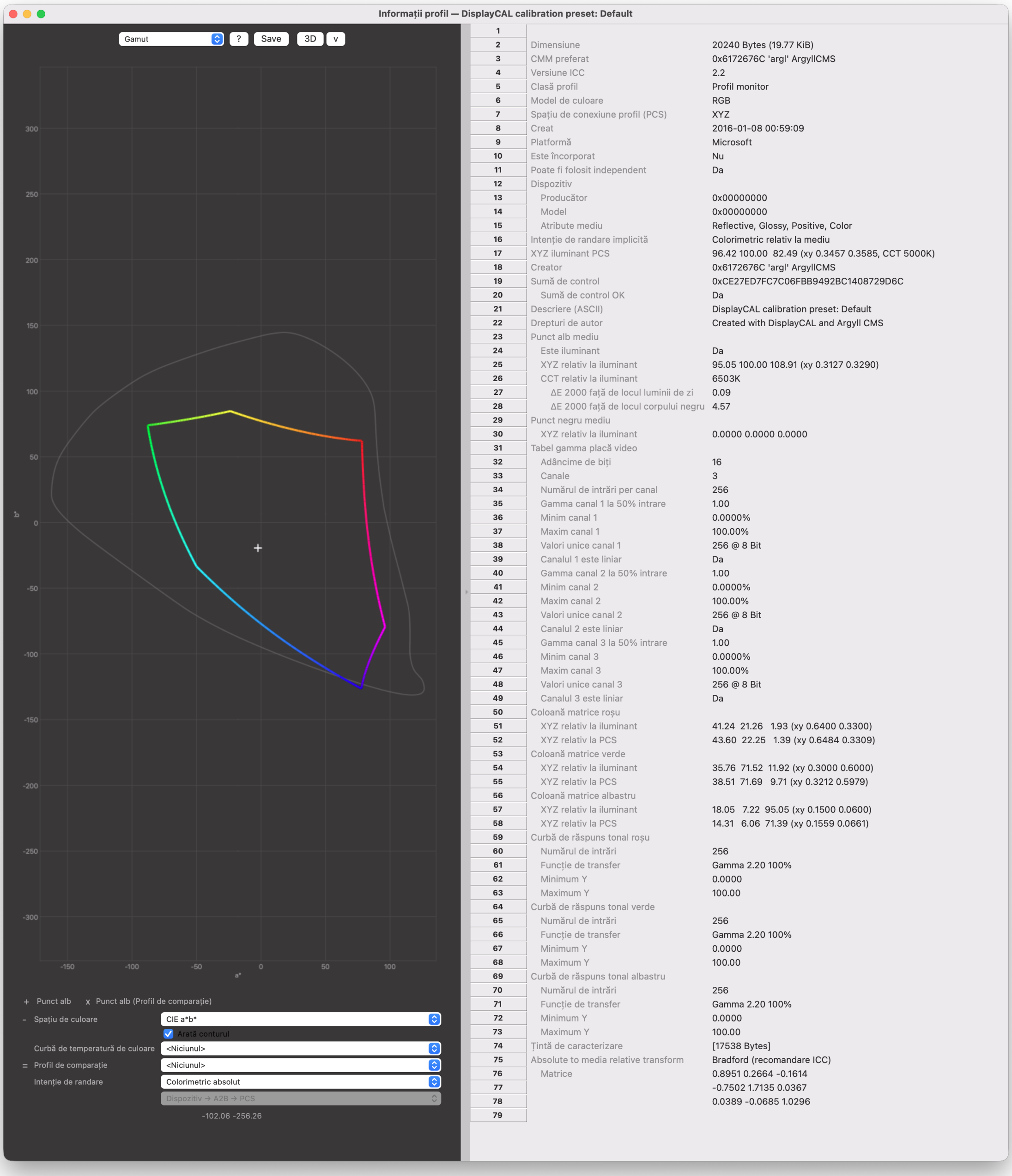
Create 3D LUT (standalone)



The standalone 3D LUT creation tool, for converting between colorspaces without going through a full display-calibration workflow.

The same 3D LUT generation logic as the "3D LUT" tab, but run independently of any display calibration workflow — useful for converting between two arbitrary profiles/colorspaces.

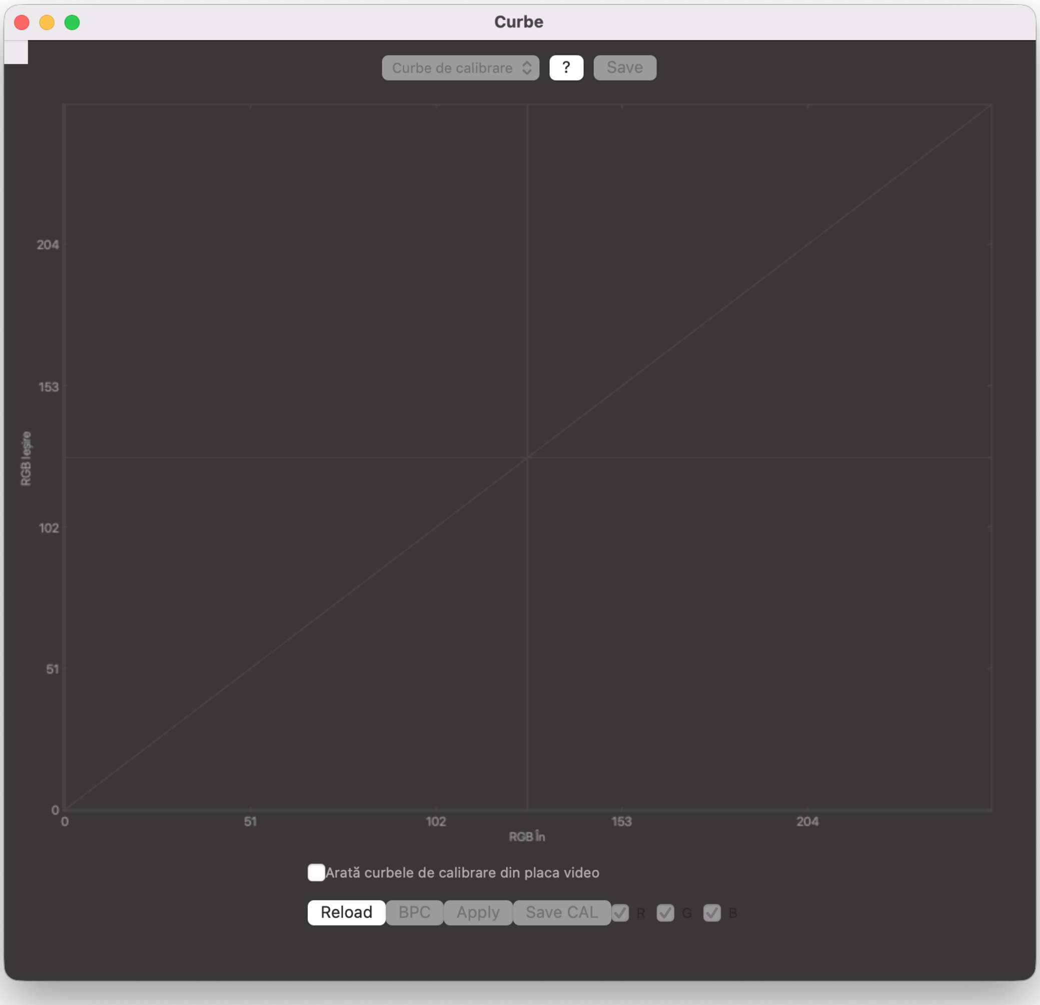
Profile Info



The Profile Info window — complete information about an ICC profile, plus a graphical representation of its gamut.

Opens any ICC profile (created by DisplayCAL-CG or from another source) and shows all the information it contains — white point, tone curve, color primaries — plus a 3D graphical representation of the covered color gamut.

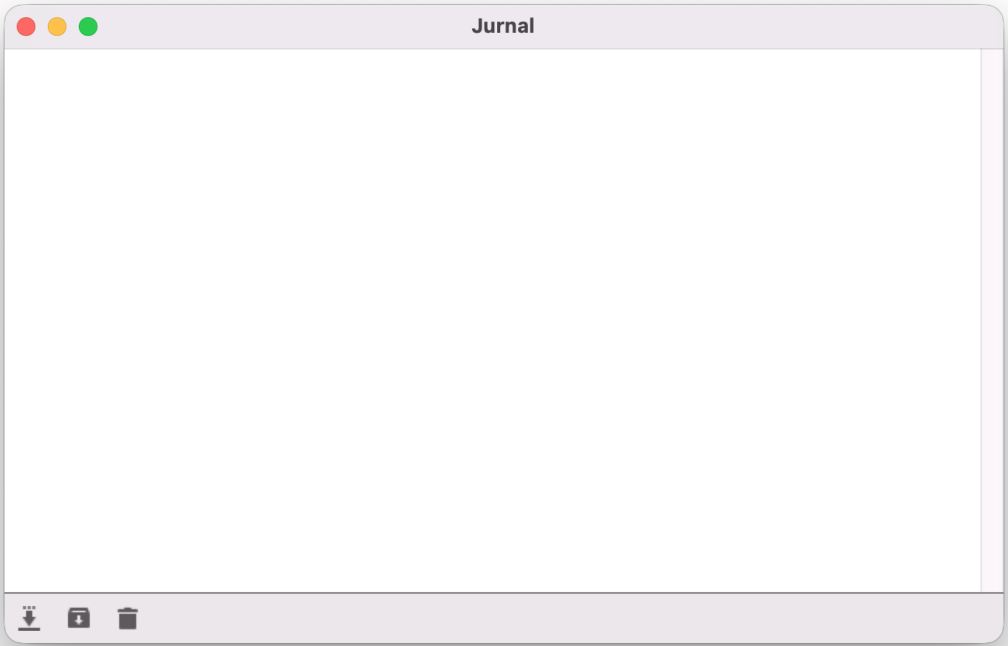
Curves



The Curves window — a visualization of the calibration curves (vcgt) currently loaded into the video card.

Graphically shows the calibration curves (VCGT — Video Card Gamma Table) currently loaded into the video card — useful for a quick visual check of what calibration is active right now.

Log



The Log window — the detailed technical log of the ArgyllCMS operations behind the application.

The detailed technical log of the ArgyllCMS commands run behind the scenes by the app — mainly useful for troubleshooting if something isn't working as expected and you want to understand exactly what happened.

9. GPLv3 license & optional support

> **100% free, forever**
> DisplayCAL-CG is free software, licensed under GPLv3 — fully functional from day one, no activation, no time-limited trial, no feature locked behind a payment. You can install, use and redistribute the application freely, subject to the included GPLv3 license terms (LICENSE.txt).

DisplayCAL-CG is built on the open-source work of DisplayCAL (Florian Höch) and its community continuators — full credits remain visible in the app (Help → About).

If the app has been useful to you, an optional support message appears occasionally in the application — it's purely informational, never a requirement to use any feature.

10. Updates

DisplayCAL-CG automatically checks at startup whether a newer version is available on the download page. You can also check manually from the Help menu.

1. If a new-version notification appears, click the link in it — it takes you straight to the download page with the latest version.
2. Download the new package (.pkg on Mac, .exe on Windows).
3. Install it over the current version, exactly as with the first install (see the "Installation" chapter) — your settings and already-created profiles remain untouched.
4. Restart the application after installing.